

Representation Beats Reasoning: Zero-Shot LLM Forecasting of 14-Day Adverse Events in Lung-Cancer Patients from Fitbit and EMA

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Abstract—Patients on tyrosine-kinase-inhibitor (TKI) therapy for lung cancer experience frequent adverse events (AEs) between clinic visits, while their Fitbit and ecological-momentary-assessment (EMA) data go unread. We ask whether a zero-shot large language model (LLM), given a 14-day window of this passive data plus an FDA-derived drug-specific AE profile, can forecast new AE onset in the next 14 days. On 30 patients and 291 leave-one-patient-out windows (13.4% positive), our best zero-shot configuration (Claude Sonnet with a clinical narrative timeline, drug-keyed retrieval, and sleep features) attains AUROC 0.623 [95% CI 0.536–0.699], statistically indistinguishable from the strongest supervised baseline (LightGBM, 0.620 [0.529–0.711]). LightGBM is in fact slightly ahead on AUPRC and per-patient AUROC. The defensible claim is therefore that an LLM with the right representation and retrieval *matches*, but does not beat, a tuned supervised model — with no training data. A clean three-step ablation isolates the gains: JSON→narrative +0.030, drug retrieval +0.035, sleep +0.028; adding few-shot or chain-of-thought removes 0.005. A leakage audit shows that stripping the CTCAE-aligned symptom EMA drops AUROC by only 0.033 but doubles mean predicted risk and triples Brier score: the symptom EMA is a calibration anchor, not the discrimination source. Per-patient AUROC is 0.46, indicating that the headline metric is driven by cross-patient prevalence differences — the contribution is cohort-level risk stratification, not within-patient early warning.

Index Terms—adverse event forecasting, large language model, retrieval-augmented generation, wearable sensing, lung cancer, calibration, zero-shot prediction.

I. INTRODUCTION

Targeted therapy with TKIs is the standard of care for EGFR- and ALK-mutated non-small-cell lung cancer. The drugs are effective, but their toxicity — diarrhea, rash, fatigue, musculoskeletal pain, and rare pneumonitis — forces dose reductions and shortens treatment duration. Clinic visits are spaced weeks apart, so AEs onset between visits and are typically caught only after symptoms are severe. Trial evidence from Basch et al. [1] showed that systematic patient-reported symptom monitoring improves overall survival, establishing that earlier AE detection has measurable clinical value. Ac-

tivity from consumer wearables has likewise been associated with clinical outcomes in advanced cancer [2].

Recent work has shown that large language models can encode and apply clinical knowledge competently [3], [4] and, more directly relevant here, can read aggregated wearable data and output health predictions [5], [6]. The specific gap our work addresses is whether an LLM can produce a useful forecasting probability over a 14-day horizon on a small oncology cohort where supervised models have very little training data to work with. We target the simplest and most reproducible recipe: a frozen, general-purpose LLM accessed zero-shot, with two engineered components — a clinical-style *narrative* representation of the 14-day window and a deterministic, drug-keyed retrieval [7] over an FDA-derived AE profile.

We pose three controlled sub-questions. RQ1 asks whether a narrative timeline beats a structured JSON summary of the same window. RQ2 asks whether injecting an FDA-derived drug-specific AE profile improves discrimination above sensing data alone. RQ3 asks whether few-shot exemplars [8] and chain-of-thought reasoning [9] add anything once the representation and retrieval choices are in place. A post-hoc audit then asks how much of the apparent signal is contributed by the CTCAE-aligned symptom EMA — the most label-adjacent feature — and what per-patient AUROC says about the headline number.

The headline finding is one of equivalence rather than superiority. The best LLM matches a strong supervised baseline (LightGBM) within overlapping 95% confidence intervals while using no training data. *All* of the LLM-internal gains came from representation and retrieval; few-shot and chain-of-thought added nothing, consistent with recent meta-evidence that chain-of-thought helps mainly on math and symbolic tasks [10] and tends to degrade clinical-text performance [11]. The leakage audit shows that discrimination is preserved when the CTCAE-aligned symptom items are removed; what collapses is calibration. Per-patient AUROC indicates the headline is cohort-level stratification, not early warning, and we frame

the contribution accordingly.

II. DATA AND TASK

a) Cohort.: Thirty patients from the HSR210257 Lung001 V2 protocol, each on a single-agent TKI from a universe of eight drugs (osimertinib, alectinib, entrectinib, mobocertinib, and four others). Per-patient observation is approximately 130 days from enrollment.

b) Modalities.: (i) **Fitbit**: 15 daily features (steps, distance, active/sedentary minutes, resting and active heart rate, sleep duration and efficiency, REM/deep/light/awake minutes, activity bouts); (ii) **Daily symptom EMA**: ten items aligned to CTCAE thresholds (fatigue, nausea, edema, rash, diarrhea, pain, appetite, mood, sleep quality, dyspnea); (iii) **Weekly EMA** (6 items: missed doses, stress, concentration) and **bi-weekly EMA** (22 items: psych, pain, social functioning), forward-filled within their intervals; (iv) the clinical **AE log** (140 AEs in 16 categories), drug table, and demographics.

A subtle but important data point: the daily symptom EMA items are phrased after the CTCAE grading manual, so a positive report is functionally a patient-reported AE at the CTCAE threshold for that symptom. The symptom EMA is therefore not a fully independent predictor — it overlaps semantically with the clinician-recorded AE label. We retain it as a feature and audit its role in Section V.

c) Task.: Sliding-window binary forecasting with look-back $w = 14$ days, stride $s = 14$ days (non-overlapping), horizon $h = 14$ days, and label $y = 1$ iff any new AE onset occurs within the next 14 days, regardless of grade or category. We initially used $s = 1$ but discarded it: adjacent windows share 13 of 14 days and that temporal correlation artificially inflates LOPO AUROC. With $s = 14$, the dataset has **291 evaluation instances, 39 positive (13.4% base rate)**.

d) Evaluation.: Leave-one-patient-out (LOPO): for each patient, all of their instances are held out. The LLM is zero-shot, so there is no training in the supervised sense, but supervised baselines are trained on the other 29 patients per split. Primary metrics are AUROC and AUPRC; we also report Brier score, sensitivity at 80% specificity, and per-patient AUROC (the mean and median across patients with both classes present, $n = 17$). Confidence intervals are 95% percentile bootstrap ($n_{\text{boot}} = 1000$).

III. METHOD

A. Pipeline

The system has five stages: (1) raw streams (Fitbit, EMA, AE log); (2) instance construction at $w=s=h=14$ days; (3) representation and retrieval — a narrative timeline plus a drug-keyed retrieval block; (4) zero-shot inference via Claude Sonnet through the `claude --print CLI`; (5) evaluation against classical and gradient-boosted ML baselines under identical LOPO splits. Predictions are cached as JSONL keyed by (`prompt_version`, `instance_id`), so prompt edits do not re-cost successful calls.

Structured JSON input:

```
{
  "total_steps": {
    "mean": 3621,
    "trend_slope": -430.7,
    "last_value": 1273
  }, ...
}
```

Narrative timeline input (same window):

```
Patient: 68F, EGFR+, osimertinib 80mg QD, on TKI 6 months.
[Activity] Steps 4717 -> 1273 (-73%)
          Trend: -431 steps/day [!] Sharp drop Day 6
[Heart Rate] Day-night gap collapsed 18 -> 4 bpm
[Sleep] Duration 6h 10m (-45m), efficiency 81% (-8 pp)
[Symptoms] Fatigue 2 -> 4 (last 3 days), nausea 1, rash 0
[AE context] no prior AE; on TKI 6 months
```

Fig. 1. The narrative is itself the feature-engineering step: we output natural language instead of a feature vector, so the LLM reasons over findings rather than over numbers.

Osimertinib (Tagrisso) -- EGFR TKI

Common AEs: diarrhea 47%, rash 46%, MSK pain 38%,
nail toxicity 34%, dry skin 32%

Serious: ILD/pneumonitis 4% (0.4% fatal), QTc

Onset: skin reactions ~1 month, ILD median ~3 months

Fig. 2. Drug profile injected into every prompt for an osimertinib patient. The retriever is a deterministic exact-match lookup on the patient’s drug name — intentionally the smallest auditable form of RAG [7], distinct from learned dense retrieval such as Self-RAG [13]. Auditability matters more than retrieval sophistication for medical decision support.

B. Narrative timeline (RQ1)

The natural way to give an LLM a 14-day multi-modal window is structured JSON — per-day arrays and aggregate statistics. The problem is that LLMs are pretrained on clinical notes and case reports, not on medical JSON schemas. Asking the model to interpret a JSON object is asking it to do a translation step that has nothing to do with the clinical question. We invert the pattern: we pre-compute the derived quantities a clinician would care about — percent change, trend slope, sharp-change flags, day–night heart-rate gap, sleep-efficiency dips — and render them as a clinical-style progress note with arrows and warning markers as salience cues. Fig. 1 shows the same window in both forms.

C. Drug-aware retrieval (RQ2)

The eight TKIs in our cohort have substantially different AE profiles. Drug-specific AE rates are documented in FDA prescribing information but are not reliably recoverable from Claude’s parametric memory — free-form recall of, e.g., osimertinib pneumonitis rates is inconsistent across runs. We externalize the knowledge into a small, curated KB: for each of the eight drugs we queried the OpenFDA drug-label endpoint, parsed the structured AE-rate and warnings sections, and produced a compact dict. For osimertinib (Tagrisso), the registrational FLAURA trial reports similar rates [12]. An example block is shown in Fig. 2.

D. Reasoning prompts (RQ3)

On top of the best representation-and-retrieval stack (V2-RAG-Sleep), we test two standard prompt-engineering moves.

V3-Fewshot prepends three example windows with outcome labels only (no reasoning chains) [8]. *V3-CoT* adds three exemplars and a six-step scaffold instructing the model to review activity and sleep, heart-rate autonomies, symptoms, drug match, time-on-therapy, and integrate [9].

IV. EXPERIMENTS

A. Main results

All variants are evaluated on the same 291 LOPO instances. Classical baselines use the same aggregate-feature vector as the JSON variant. Gradient-boosted baselines (“Strong-ML”) use a 420-feature engineered set averaged over five seeds. Table I reports AUROC and AUPRC with 95% percentile-bootstrap CIs, Brier score, and per-patient AUROC.

B. What the table actually says

The two top-line numbers are V2-RAG-Sleep at AUROC 0.623 [0.536, 0.699] and LightGBM at 0.620 [0.529, 0.711]. Their confidence intervals overlap almost completely. LightGBM is slightly ahead on AUPRC (0.204 vs. 0.187) and noticeably ahead on per-patient AUROC (0.562 vs. 0.459). The defensible reading is:

A zero-shot LLM with the right representation and drug-specific retrieval matches a tuned supervised baseline on this task — without ever seeing a training label — but it does not beat it.

This is a useful finding precisely because it is bounded. In small-cohort clinical settings where no labeled corpus exists to train on, the LLM-with-retrieval recipe is competitive out of the box; in settings where a strong supervised model can be trained, the LLM is not an obvious replacement.

C. Decomposing the LLM-internal gains

Table II isolates where the LLM gains came from. Representation and retrieval together moved AUROC from 0.529 to 0.623 (+0.094 cumulative). Adding reasoning prompts on top removed ≈ 0.005 . Few-shot also collapsed calibration: with two of three exemplars positive (67% vs. a 13% cohort base rate), predicted mean risk inflated to 0.569 and Brier reached 0.317. This is the predicted in-context-learning failure mode — demonstrations communicate the label distribution more than the reasoning content [14].

V. LEAKAGE AUDIT

Because the daily symptom EMA is CTCAE-aligned (Section III-B), a natural worry is that the LLM’s V2-RAG-Sleep result reflects retrospective concordance with patient self-reports rather than prospective forecasting. We test this with a controlled comparison. *Variant A* is V2-RAG-Sleep as reported in Table I. *Variant B* is the same prompt and same model with the [Symptoms] block removed and all inline symptom values stripped from the last-3-days table; Fitbit, weekly non-symptom EMA, biweekly psych/pain/social items, drug retrieval, and demographics are kept. Variant B had eight CLI timeouts, so the comparison is on the 283 matched instances common to A and B (37 positive, 13.1%).

The initial hypothesis — that stripping the most label-adjacent input would collapse discrimination — is rejected. AUROC drops by only 0.033; the rank correlation between A and B is 0.774. What collapses is calibration: mean predicted risk roughly doubles, Brier triples, and at the natural threshold of 0.20 on Variant B, recall reaches 1.000 with precision equal to the base rate. The mechanism is the opposite of the standard leakage worry. The LLM uses *absence of symptoms* as a low-risk anchor; without that anchor it defaults to high risk across the board. Discrimination ranking is supplied by Fitbit, non-symptom EMA, drug retrieval, and demographics — none of which are label-adjacent.

A second finding from the audit is independent of the manipulation: the per-patient AUROC of the headline configuration is 0.459 ± 0.229 , with only 7 of 17 patients with both classes present achieving a within-patient AUROC above 0.5. A near-random per-patient AUROC alongside an overall AUROC of 0.62 indicates that the headline metric is driven by cross-patient prevalence differences rather than within-patient time discrimination. We therefore reframe the contribution: we do not claim “early warning” or “within-patient forecasting”; we claim *cohort-level risk stratification* — given the patient’s overall profile, which patients sit in the high-AE-burden subgroup. This is the small-cohort prognostic pitfall flagged in the methodological literature [15].

VI. DISCUSSION

a) Representation > reasoning.: The three research questions were designed to isolate three levers — representation, retrieval, reasoning prompts. The first two contributed +0.030 and +0.035 AUROC; the third contributed nothing measurable, and few-shot exemplars actively damaged calibration in the predicted way [14]. This is consistent with concurrent meta-analyses that chain-of-thought helps mainly on math and symbolic tasks [10] and tends to degrade clinical-text understanding [11]. On sensor-grounded health prediction, choosing what to put in front of the model matters more than telling the model how to reason.

b) What “zero-shot” actually buys.: The LLM’s no-training setting carries a heavy implicit prior — it has absorbed the medical literature including TKI AE descriptions and clinical-note conventions. LightGBM does not. Reading our result charitably, medical pretraining is roughly as useful as ~ 290 LOPO training instances of aggregated Fitbit and EMA features. Reading it skeptically, the comparison is between an extensively pretrained model and an untrained one. Both readings are correct; the first explains why the approach is promising in small-cohort oncology settings, the second explains why this is not yet an LLM-beats-ML claim.

c) Calibration.: The mean predicted risk of V2-RAG-Sleep is 0.251 vs. a true base rate of 0.134, so the model is $\approx 2\times$ too high. Brier score is 0.134. We did not apply post-hoc calibration [16] because LOPO does not yield a clean held-out calibration set; a patient-stratified calibration split is the natural next step.

TABLE I
MAIN RESULTS ON 291 LOPO INSTANCES (39 POSITIVE, 13.4% BASE RATE). CONFIDENCE INTERVALS ARE 95% PERCENTILE BOOTSTRAP,
 $n_{\text{BOOT}} = 1000$. LLM IS CLAUDE SONNET, ZERO-SHOT.

Family	Model	AUROC (95% CI)	AUPRC (95% CI)	Brier	Per-pt AUROC
LLM zero-shot	V1-JSON	0.529 [0.442, 0.628]	0.149 [0.102, 0.230]	0.124	0.366
	V2-Narrative	0.560 [0.484, 0.640]	0.156 [0.110, 0.214]	0.123	0.387
	V2-RAG	0.595 [0.506, 0.687]	0.183 [0.122, 0.288]	0.134	0.417
	V2-RAG-Sleep	0.623 [0.536, 0.699]	0.187 [0.124, 0.282]	0.134	0.459
	V3-Fewshot	0.617 [0.523, 0.698]	0.185 [0.126, 0.286]	0.317	0.434
	V3-CoT	0.618 [0.525, 0.704]	0.181 [0.124, 0.277]	0.157	0.442
Strong ML	LightGBM	0.620 [0.529, 0.711]	0.204 [0.138, 0.322]	0.130	0.562
	Random Forest	0.578 [0.496, 0.666]	0.167 [0.114, 0.266]	0.137	0.525
	XGBoost	0.556 [0.451, 0.655]	0.189 [0.121, 0.300]	0.134	0.505
	HistGB	0.555 [0.445, 0.659]	0.172 [0.116, 0.270]	0.138	0.528
Reference	Always-predict-0	0.500	0.134	—	—

TABLE II
LLM-INTERNAL DECOMPOSITION. ALL GAINS CAME FROM REPRESENTATION AND RETRIEVAL; REASONING PROMPTS CONTRIBUTED NOTHING.

Transition	Δ AUROC
V1-JSON $\rightarrow$ V2-Narrative	+0.030
V2-Narrative $\rightarrow$ V2-RAG	+0.035
V2-RAG $\rightarrow$ V2-RAG-Sleep	+0.028
V2-RAG-Sleep $\rightarrow$ V3-CoT	−0.005
V2-RAG-Sleep $\rightarrow$ V3-Fewshot	−0.006
V1-JSON $\rightarrow$ V2-RAG-Sleep	+0.094

TABLE III
LEAKAGE AUDIT. 283 MATCHED INSTANCES. SPEARMAN RANK CORRELATION A VS. B = 0.774.

Metric	A: with sym	B: no sym	Δ
AUROC	0.624	0.591	−0.033
AUPRC	0.186	0.182	−0.004
Brier	0.131	0.310	+0.179
Mean predicted risk	0.249	0.544	+0.295
Per-pt AUROC mean	0.459	0.354	−0.105
Per-pt AUROC median	0.438	0.333	−0.105
Per-pt > 0.5	7/17	3/17	−4

d) *Limitations.*: (i) Thirty patients, 291 windows, 39 positives: both LLM and ML CIs have width ~ 0.17 . (ii) Within-patient discrimination is not established; per-patient AUROC is near random. (iii) Calibration is not corrected. (iv) Single site, single protocol. (v) Drug imbalance: some TKIs appear in only one or two patients, so the drug-retrieval benefit is tested in aggregate, not per-drug. (vi) Inference is via the Claude CLI on a subscription tier; the cache is exactly reproducible, but re-inference is only approximately so if the underlying model is silently revised. (vii) Labels are grade- and category-agnostic; a clinical deployment would want both.

e) *Future work.*: Personal-baseline features rendered into the narrative as deviations from each patient’s own range

directly address the per-patient AUROC limitation. Beyond static drug profiles, dynamic case retrieval — retrieving similar past patient trajectories from the cohort as in-context examples — is the natural extension, and the agentic recipe demonstrated by PHIA [17] suggests a route. A complementary direction is foundation models that learn directly over numeric sensor time series [18], distinct from our narrative-and-retrieval pattern.

VII. CONCLUSION

A zero-shot LLM, given a clinical-style narrative timeline of a 14-day Fitbit and EMA window plus a deterministic, drug-keyed retrieval over an FDA-derived AE knowledge base, attains AUROC 0.623 on a 30-patient TKI lung-cancer cohort and matches a strong supervised baseline (LightGBM 0.620) without using any training data. Few-shot prompting and chain-of-thought reasoning, on top of the representation-and-retrieval stack, contributed nothing measurable. A controlled leakage audit shows that the CTC AE-aligned symptom EMA functions as a calibration anchor rather than the discrimination source, and per-patient AUROC near 0.46 reframes the contribution from “early warning” to “cohort-level risk stratification.” The simplest summary: on clinical sensor-grounded forecasting, choosing what to put in front of the model matters far more than telling the model how to reason, and per-patient evaluation is the metric that separates genuine forecasting from prevalence-driven stratification.

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